













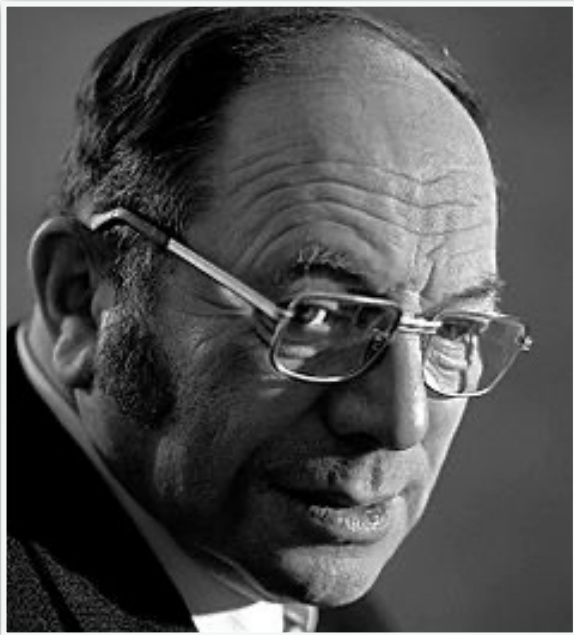










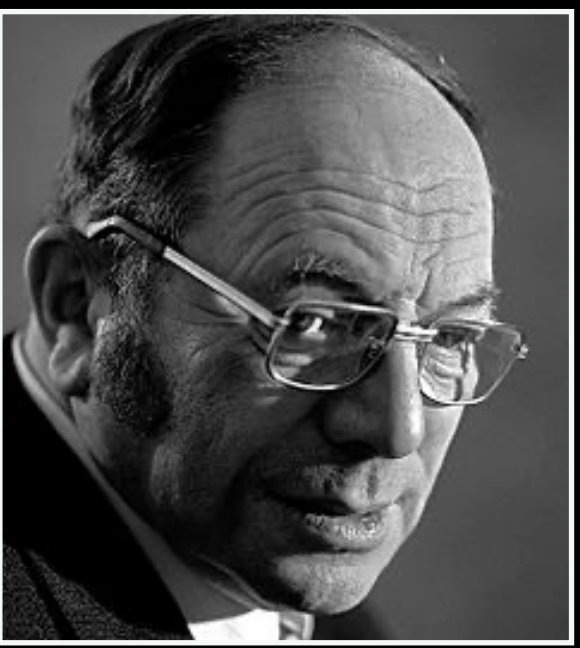




Takeaway: The Kantorovich formulation trades off flexibility/well-posedness for complexity



# Monge vs Kantorovich



- **Existence:** Monge not always solvable; Kantorovich always is (feasible set is bounded, objective is linear).
- **Uniqueness:** as with LPs, Kantorovich can have multiple optimal solutions
- **Feasible region:** Monge non convex, Kantorovich convex (convex hull)
- Monge is asymmetric ( $T$  is a valid pushforward from  $a$  and  $b \not\Rightarrow T^{-1}$  is valid), Kantorovich is symmetric:  $\pi \in \Pi(\mathbf{a}, \mathbf{b}) \iff \pi^\top \in \Pi(\mathbf{b}, \mathbf{a})$
- **Complexity:**
  - Monge = assignment problem over  $n$  vars:  $O(n^3)$
  - Kantorovich = LP over  $n \times n$  couplings:  $O(n^6)$  (naive), or  $O(n^3)$  to  $O(n^4)$  with specialized solvers
  - Typically LP algos slower in practice than assignment solvers

Takeaway: The Kantorovich formulation trades off flexibility/well-posedness for complexity

# Monge vs Kantorovich

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